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FlowEdit: Inversion-Free Text-Based Editing Using Pre-Trained Flow Models

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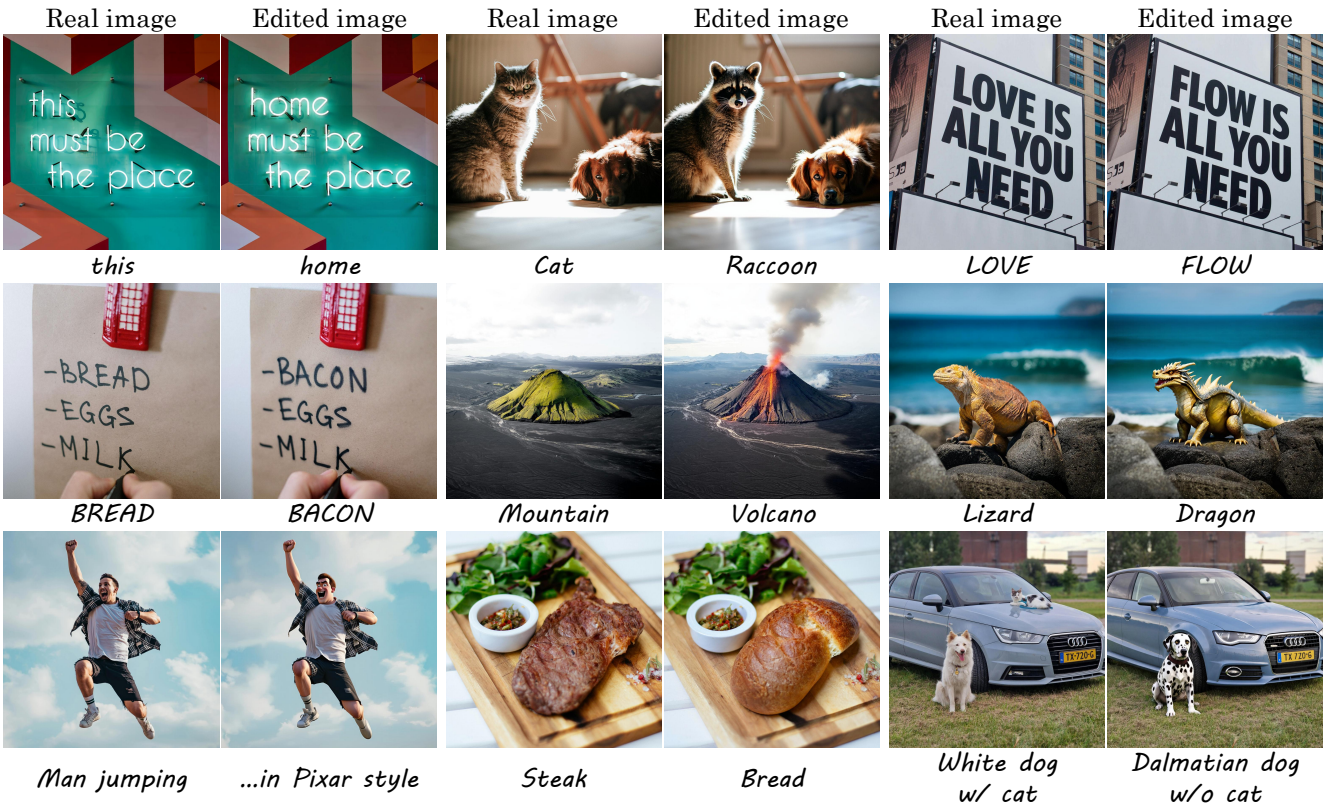


Figure 1. **FlowEdit.** We present an inversion-free, optimization-free and model agnostic method for text-based image editing using pre-trained flow models. As opposed to the editing-by-inversion paradigm, FlowEdit constructs an ODE that directly maps the source image distribution to the target image distribution (corresponding to the source and target prompts). This ODE achieves a lower transport cost and thus leads to better structure preservation, achieving state of the art results on complex editing tasks. From left to right, top to bottom, the first five images were obtained with FLUX and the rest with Stable Diffusion 3. Text indicates changes in the prompts.

Abstract

Editing real images using a pre-trained text-to-image (T2I) diffusion/flow model often involves inverting the image into its corresponding noise map. However, inversion by itself is typically insufficient for obtaining satisfactory results, and therefore many methods additionally intervene in the sampling process. Such methods achieve improved results but are not seamlessly transferable between model architectures. Here, we introduce FlowEdit, a text-based editing method for pre-trained T2I flow models, which is inversion-free, optimization-free and model agnostic. Our method constructs an ODE that directly maps between the source and target distributions (corresponding to the source and target text prompts) and achieves a lower transport cost than the inversion approach. This leads to state-of-the-art results, as we illustrate with Stable Diffusion 3 and FLUX. Code and examples are available on the project's webpage.

1. Introduction

Diffusion and flow models generate samples through an iterative process, which starts from pure noise and employs denoising-like steps. Beyond data generation, many methods employ pre-trained diffusion/flow models for editing real signals [5, 7, 12, 14–16, 19, 20, 23, 29, 30, 37, 44, 48]. While some techniques rely on test-time optimization [15, 20], many methods aim for optimization-free approaches. A common first step in the latter category of methods is *inversion*, *i.e*. extracting the initial noise vector that presumably generates the signal one wishes to edit. This noise vector is then used to generate the edited signal by injecting a different condition to the model, such as a modified text prompt [19, 32, 45, 47].

In the context of image editing, many works noted that this editing-by-inversion paradigm often leads to insufficient fidelity to the source image (see *e.g*. [19]). To overcome this, some methods attempt to reduce the errors incurred in the inversion process [11, 32]. However, this rarely eliminates the problem as even when editing generated images, for which the initial noise is precisely known, editing-by-inversion commonly fails to preserve structure. Other methods intervene in the sampling process by injecting internal model representations (*e.g*. attention maps) recorded during the inversion process [6, 16, 44]. These methods achieve improved fidelity but are not easily transferable across model architectures and sampling schemes.

In this work, we present FlowEdit, a text-based editing method for pre-trained text-to-image (T2I) flow models, which breaks away from the editing-by-inversion paradigm. FlowEdit is optimization-free and model agnostic, making it easily transferable between models. Rather than mapping the image to noise and the noise back to an edited image, it constructs a direct path between the source and target distributions (corresponding to the source and target text prompts). This path is shorter than that achieved by inversion, and thus maintains better fidelity to the source image. We evaluate FlowEdit on diverse edits using FLUX [24] and Stable Diffusion 3 (SD3) [10], and show that it achieves state-of-the-art results across the board (see Fig. 1).

Our method is motivated by a novel reinterpretation of editing-by-inversion as a direct path between the source and target distributions. Armed with this observation, we derive an alternative direct path that has a lower transport cost, as we show on synthetic examples.

2. Related work

To edit a real image using a pre-trained diffusion model, some methods use optimization over the image itself [15, 21, 22, 33]. These methods utilize the generative prior of a T2I diffusion model as a loss, which they optimize to push the image to comply with the user-provided prompt. Recently, Yang et al. [49] suggested a similar method for flow models. These methods are resource intensive at test-time.

Many zero-shot editing methods do not require optimization. The first step of most of these methods is image-to-noise inversion [5, 6, 8, 16, 19, 40, 43–45, 47]. However, the noise map obtained through naive inversion [41, 47] is generally unsuitable for effective editing [16, 19, 32]. This is often attributed to the inaccurate inversion process, leading to attempts to improve the inversion accuracy [13, 31, 32, 45]. Yet, even exact inversion on synthetic data leads to unsatisfactory editing [19]. To address this, many methods extract during the inversion stage structural information implicitly encoded within the model architecture. They then inject it in the sampling process to achieve better structure preservation [5, 6, 16, 34, 44]. These methods are typically tailored for specific model architectures and sampling methods, limiting their transferability to new settings, such as to flow models [10, 24]. Several recent methods were proposed for editing with flow models [3, 39, 46]. However, these methods rely on inversion and in some cases also on feature injection, and thus suffer from the same limitations as their diffusion-model counterparts.

Unlike previous works, our method does not rely on inversion. It maps from source to target distributions without traversing through the Gaussian distribution. It also avoids optimization and does not intervene in the model internals, making it easily adaptable to new models.

3. Preliminaries

3.1. Rectified Flow models

Generative flow models attempt to construct a transportation between the distributions of two random vectors, X_0 and X_1, defined by an ordinary differential equation (ODE) over time $t \in [0, 1]$,

$$dZ_t = V(Z_t, t)\, dt. \tag{1}$$

Here, V is a time-dependent velocity field, usually parameterized by a neural network, which has the property that if the boundary condition at $t = 1$ is $Z_1 = X_1$, then Z_0 is distributed like X_0. It is common to choose $X_1 \sim \mathcal{N}(0, \boldsymbol{I})$, which allows to easily draw samples from the distribution of X_0. This is done by initializing the ODE at $t = 1$ with a sample of a standard Gaussian vector, and numerically solving the ODE backwards in time to obtain a sample Z_0 from the distribution of X_0 at time $t = 0$.

Rectified flows [2, 26, 28] are a particular type of flow models, which are trained such that the marginal at time t corresponds to a linear interpolation between X_0 and X_1,

$$Z_t \sim (1 - t)X_0 + tX_1. \tag{2}$$

Rectified flows have the desired property that their sampling paths are relatively straight. This allows using a small number of discretization steps when solving the ODE.

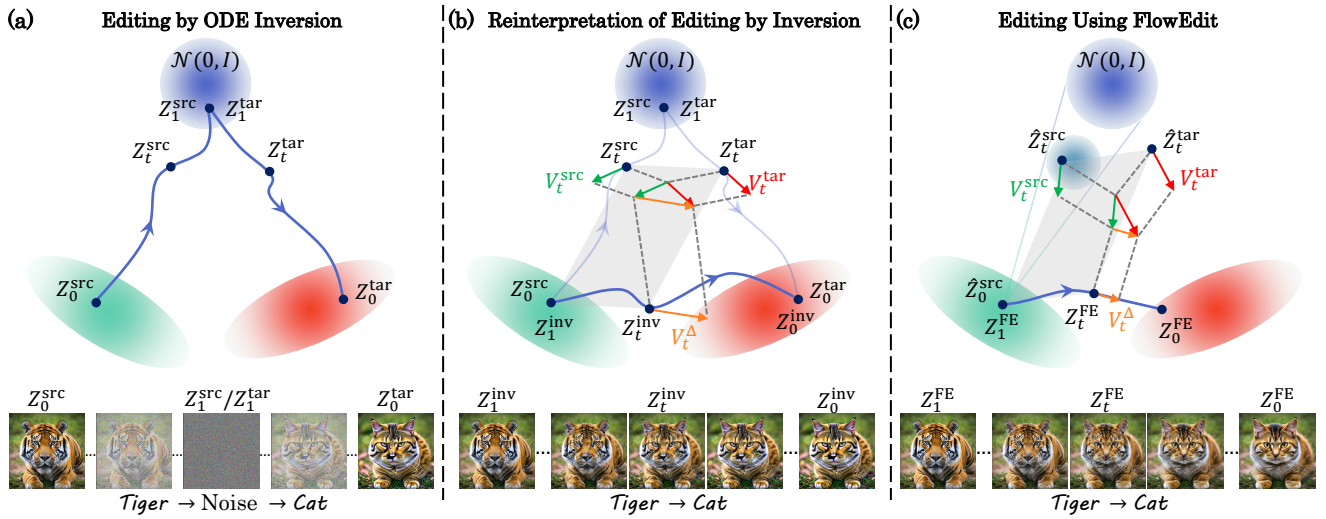


Figure 2. **Editing by inversion vs. FlowEdit.** **(a)** In inversion based editing, the source image Z_0^{src} is first mapped to the noise space by solving the forward ODE conditioned on the source prompt (left path). Then, the extracted noise is used to solve the reverse ODE conditioned on the target prompt to obtain Z_0^{tar} (right path). The images at the bottom visualize this transition. **(b)** We reinterpret inversion as a direct path between the source and target distributions (bottom path). This is done by using the velocities calculated during the inversion and sampling (green and red arrows) to calculate an editing direction (orange arrow) that drives the evolution of the direct path Z_t^{inv} through an ODE. The resulting path is noise-free, as demonstrated by the images at the bottom. **(c)** FlowEdit traverses a *shorter* direct path, Z_t^{FE}, without relying on inversion. At each timestep, we directly add random noise to $\hat{Z}_0^{\text{src}}$ to obtain $\hat{Z}_t^{\text{src}}$ and use that direction to create $\hat{Z}_t^{\text{tar}}$ from Z_t^{FE}(gray parallelogram). We then calculate the corresponding velocities and average over multiple realizations (not shown in the figure) to obtain the next ODE step (orange arrow). The images at the bottom demonstrate our noise-free path.

Text-to-image flow models employ a velocity field $V(X_t, t, C)$ that depends on a text prompt C. Such models are trained on pairs of text and image data, (C, X_0), so as to allow sampling from the conditional distribution of $X_0|C$.

3.2. Image editing using ODE inversion

Suppose we are given a real image, X^{src}, which we want to edit by providing a text prompt c_{src} describing the image and a text prompt c_{tar} describing the desired edit. A common approach to do so is by using a pre-trained text-conditioned diffusion/flow model and employing *inversion*. Specifically, let us denote the text conditioned velocities as $V^{\text{src}}(X_t, t) \triangleq V(X_t, t, c_{\text{src}})$ and $V^{\text{tar}}(X_t, t) \triangleq V(X_t, t, c_{\text{tar}})$. Methods relying on inversion start by extracting the initial noise map corresponding to the source image. This is done by traversing the forward process defined by the ODE

$$dZ_t^{\text{src}} = V^{\text{src}}(Z_t^{\text{src}}, t)\, dt, \tag{3}$$

starting at $t = 0$ from the source image, $Z_0^{\text{src}} = X^{\text{src}}$, and reaching a noise map Z_1^{src} at $t = 1$. The process Z_t^{src} is the path shown on the left of Fig. 2a. Then, the reverse ODE,

$$dZ_t^{\text{tar}} = V^{\text{tar}}(Z_t^{\text{tar}}, t)\, dt, \tag{4}$$

is solved backward in time, starting from $t = 1$ with the extracted noise, $Z_1^{\text{tar}} = Z_1^{\text{src}}$, and reaching an edited image Z_0^{tar} at $t = 0$. This is the path shown on the right of Fig. 2a.

Editing by inversion tends to produce unsatisfactory results, and is thus usually accompanied by feature map injection. While the sub-optimal results are commonly blamed on the inaccurate discretization of the ODE, they are also encountered when editing generated images and using their ground-truth initial noise maps [19]. Our method retains the simplicity of editing by inversion and achieves state of the art results without any intervention in the model internals.

4. Reinterpretation of editing by inversion

Before presenting FlowEdit, we reinterpret the editing-by-inversion approach. This will serve to motivate our method.

Inversion based editing transports between the source and target distributions, while passing through the distribution of Gaussian noise. However, this approach can also be expressed as a direct path between the source and target distributions. Namely, we can construct a path Z_t^{inv} that starts at $t = 1$ in the source distribution and reaches the target distribution at $t = 0$. Specifically, given the forward and reverse flow trajectories, Z_t^{src} and Z_t^{tar}, we define

$$Z_t^{\text{inv}} = Z_0^{\text{src}} + Z_t^{\text{tar}} - Z_t^{\text{src}}. \tag{5}$$

Note that when going from $t = 1$ to $t = 0$ we transition from the source image, $Z_1^{\text{inv}} = Z_0^{\text{src}} = X^{\text{src}}$ (because $Z_1^{\text{tar}} = Z_1^{\text{src}}$), to the edited image, $Z_0^{\text{inv}} = Z_0^{\text{tar}}$. This path is depicted

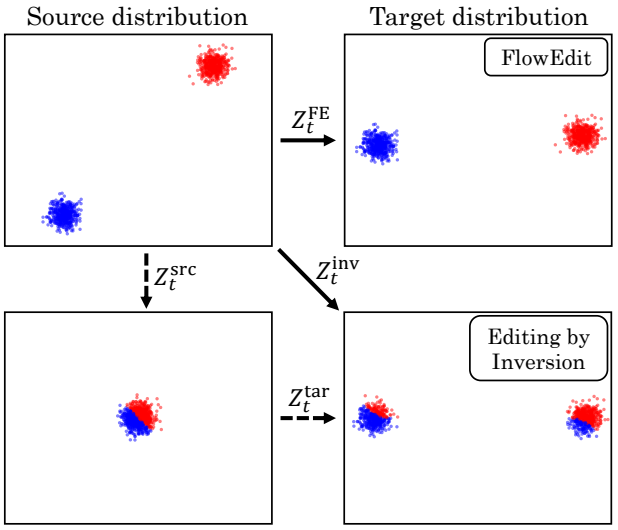


Figure 3. **Source to target pairings for editing-by-inversion and FlowEdit.** Samples from the two modes in the source distribution are colored in blue and red (top left). As can be seen in the right panes, FlowEdit results in *separated* blue and red modes, as opposed to editing-by-inversion, where the modes are *intermixed*, indicating higher transport cost in terms of MSE. This can be explained by the Gaussian in the bottom left, through which the editing-by-inversion path must traverse before reaching the target. Please see the supplementary video for animated visualizations.

at the bottom of Fig. 2b, in which the gray parallelogram visualizes the relation (5).

Let us express (5) as an ODE. Differentiating both sides of (5) and substituting (3) and (4), this relation becomes

$$dZ_t^{\text{inv}} = V_t^{\Delta}(Z_t^{\text{src}}, Z_t^{\text{tar}})\, dt, \tag{6}$$

where $V_t^{\Delta}(Z_t^{\text{src}}, Z_t^{\text{tar}}) = V^{\text{tar}}(Z_t^{\text{tar}}, t) - V^{\text{src}}(Z_t^{\text{src}}, t)$. This is visualized by the green, red, and orange arrows in Fig. 2b. Now, isolating Z_t^{tar} from (5) and substituting it into (6), we get that the direct path is the solution of the ODE

$$dZ_t^{\text{inv}} = V_t^{\Delta}(Z_t^{\text{src}}, Z_t^{\text{inv}} + Z_t^{\text{src}} - Z_0^{\text{src}})\, dt \tag{7}$$

with boundary condition $Z_1^{\text{inv}} = Z_0^{\text{src}}$ at $t = 1$.

There exist, of course, many paths connecting Z_0^{src} and Z_0^{tar}. What is so special about the path defined by (7)? It turns out that images along this path are noise-free, as shown at the bottom of Fig. 2b. The reason this happens is that the noisy images Z_t^{tar} and Z_t^{src} contain roughly the same noise constituent. Therefore, the vectors $V^{\text{tar}}(Z_t^{\text{tar}}, t)$ and $V^{\text{src}}(Z_t^{\text{src}}, t)$, which point at the source and target distributions, respectively, remove roughly the same noise component, so that the difference vector $V_t^{\Delta}(Z_t^{\text{src}}, Z_t^{\text{tar}})$ (orange arrow in Fig. 2b), encompasses the difference only between the clean image predictions.

How do images evolve along this direct path? At the beginning of the path (t close to 1), the noise level in Z_t^{src} and Z_t^{tar} is substantial, and therefore the vector $V_t^{\Delta}(Z_t^{\text{src}}, Z_t^{\text{tar}})$ captures only differences in coarse image structures. As t gets smaller, the noise level drops and higher frequency contents are unveiled. In other words, the path (7) constitutes a sort of autoregressive coarse-to-fine evolution from Z_0^{src} to Z_0^{tar}. This can be seen in the images at the bottom of Fig. 2b, where the first features that get modified are the coarse structures, and the last features to get updated are the fine textures. See App. J for additional illustrations.

5. FlowEdit

The fact that editing-by-inversion can be expressed as a direct ODE does not imply that it induces a desirable pairing between source and target samples. Figure 3 shows a simple example, where the source and target distributions are Gaussian mixtures with modes centered at $\{(-\frac{15}{\sqrt{2}}, -\frac{15}{\sqrt{2}}), (\frac{15}{\sqrt{2}}, \frac{15}{\sqrt{2}})\}$ and $\{(-15, 0), (0, 15)\}$, respectively. A good editing method should map each mode in the source distribution to its closest mode in the target distribution. In the context of image editing, this means that source images are modified as little as possible, while transporting to the target distribution. However, as seen on the bottom right of Fig. 3, this is not what happens in inversion. The reason the induced pairings look the way they look can be understood by inspecting how samples are mapped to their "initial" Gaussian noise component (bottom left).

Our goal is to construct an alternative mapping, which leads to lower distances between the source and target samples. Our method, which we coin *FlowEdit*, is illustrated on the top right of Fig. 3. As can be seen, it maps each mode in the source distribution to its nearest mode in the target distribution. Compared to editing-by-inversion, FlowEdit achieves less than twice as low a transportation cost, measured by the average squared distance between source samples and their paired target samples. We emphasize that our method is based on a heuristic, which is not guaranteed to precisely map to the target distribution. In particular, it does not necessarily coincide with the optimal transport mapping. Yet, as we will see in the context of images, it achieves state-of-the-art results in terms of structure preservation (*i.e.* small transportation cost) while adhering to the target text prompt (generating samples in the target distribution).

How can we depart from the pairings dictated by inversion? Our key idea is to use multiple random pairings, and average their associated velocity fields. More concretely, we propose to use the same ODE as (7), but to replace the inversion path Z_t^{inv} by a different random process with the same marginals. Then, at each timestep, we average the directions corresponding to different realizations of that process. Specifically, consider the alternative forward process

$$\hat{Z}_t^{\text{src}} = (1-t)Z_0^{\text{src}} + tN_t, \tag{8}$$

Algorithm 1 Simplified algorithm for FlowEdit

Input: real image X^{src}, $\{t_i\}_{i=0}^{T}$, $n_{\max}$, n_{avg}
Output: edited image X^{tar}
Init: $Z^{\text{FE}}_{t_{\max}} = X^{\text{src}}_0$
for $i = n_{\max}$ **to** 1 **do**
  $N_{t_i} \sim \mathcal{N}(0, 1)$
  $Z^{\text{src}}_{t_i} \leftarrow (1 - t_i)X^{\text{src}} + t_i N_{t_i}$
  $Z^{\text{tar}}_{t_i} \leftarrow Z^{\text{FE}}_{t_i} + Z^{\text{src}}_{t_i} - X^{\text{src}}$
  $V^{\Delta}_{t_i} \leftarrow V^{\text{tar}}(Z^{\text{tar}}_{t_i}, t_i) - V^{\text{src}}(Z^{\text{src}}_{t_i}, t_i)$
  $Z^{\text{FE}}_{t_{i-1}} \leftarrow Z^{\text{FE}}_{t_i} + (t_{i-1} - t_i)V^{\Delta}_{t_i}$
end for
Return: $Z^{\text{FE}}_0 = X^{\text{tar}}_0$

Optionally average n_{avg} samples

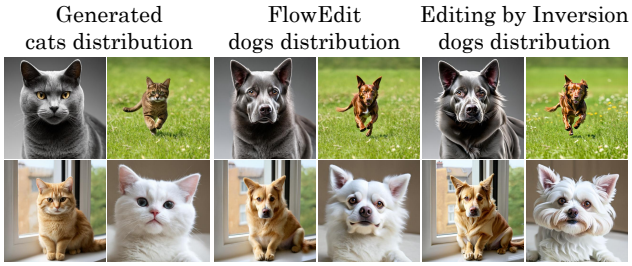


Figure 4. **Cats and dogs experiment.** We generated 1000 cat images using diverse prompts and edited them to dog images using both FlowEdit and inversion. FlowEdit outperforms editing by inversion, achieving a lower transport costs and better FID and KID scores (computed against 1000 generated dog images).

where N_t is a Gaussian process that is statistically independent of Z^{src}_0 and has marginals $N_t \sim \mathcal{N}(0, 1)$ for every $t \in [0, 1]$. We construct a path Z^{FE}_t by solving the ODE

$$dZ^{\text{FE}}_t = \mathbb{E}\left[V^{\Delta}_t(\hat{Z}^{\text{src}}_t, Z^{\text{FE}}_t + \hat{Z}^{\text{src}}_t - Z^{\text{src}}_0)\middle| Z^{\text{src}}_0\right] dt \quad (9)$$

with boundary condition $Z^{\text{FE}}_1 = Z^{\text{src}}_0$ at $t = 1$. Note that the expectation here is w.r.t. $\hat{Z}^{\text{src}}_t$ (equivalently N_t). Also note that we need not specify the covariance function of N_t as (9) depends only on the marginals of the process $\hat{Z}^{\text{src}}_t$ and not on its entire distribution law.

A schematic illustration is shown in Fig. 2c. The point $\hat{Z}^{\text{src}}_t$ on the left corresponds to a single draw of N_t. The distribution of $\hat{Z}^{\text{src}}_t$ at time t is shown in cyan. For each such draw, we calculate $\hat{Z}^{\text{tar}}_t = Z^{\text{FE}}_t + \hat{Z}^{\text{src}}_t - Z^{\text{src}}_0$, as in the editing-by-inversion ODE. This is indicated by the gray parallelogram. Now, given $\hat{Z}^{\text{src}}_t$ and $\hat{Z}^{\text{tar}}_t$, we compute the velocity field $V^{\Delta}_t(\hat{Z}^{\text{src}}_t, \hat{Z}^{\text{tar}}_t) = V^{\text{tar}}(\hat{Z}^{\text{tar}}_t, t) - V^{\text{src}}(\hat{Z}^{\text{src}}_t, t)$ to obtain an update direction. This is illustrated by the green, red, and orange arrows. We repeat this multiple times and average the resulting orange arrows (not shown in the figure) to obtain the final update direction.

5.1. Practical considerations

In practice, a discrete set of timesteps $\{t_i\}_{i=0}^{T}$ is used to drive the editing process, where T is the number of discretization steps. Additionally, the expectation in (9) is approximated by averaging n_{avg} model predictions at each timestep. As opposed to the theoretical expectation operator, when taking n_{avg} to be small, the covariance function of N_t starts playing a role. To obtain a good approximation, it is possible to exploit the averaging that naturally occurs across timesteps. We do so by choosing the covariance function of N_t to satisfy $\mathbb{E}[N_t N_s] = 0$ for every $|t - s| > \delta$, where δ is the ODE discretization step, so that the noise becomes independent across timesteps.

Note that the averaged velocity term in (9) corresponds to the difference $\mathbb{E}[V^{\text{tar}}(\hat{Z}^{\text{tar}}_t, t)|Z^{\text{src}}_0] - \mathbb{E}[V^{\text{src}}(\hat{Z}^{\text{src}}_t, t)|Z^{\text{src}}_0]$, which could be computed by sampling $\hat{Z}^{\text{tar}}_t$ and $\hat{Z}^{\text{src}}_t$ independently. However, we choose to compute them with the same noise instances, which further improves robustness to small values of n_{avg}. This is aligned with the observation of Hertz et al. [15] in the context of diffusion models, that calculating a difference between correlated noisy marginals reduces artifacts. See Sec. 5.2 for further discussion.

Similarly to [19, 30], we define an integer that determines the starting timestep for the process $0 \leq n_{\max} \leq T$, meaning that the process is initialized with $Z^{\text{FE}}_{t_{n_{\max}}} = X^{\text{src}}$. When $n_{\max} = T$, the full edit path is traversed and the strongest edit is obtained. When $n_{\max} < T$, the first $(T - n_{\max})$ timesteps are skipped, effectively shortening the edit path. This is equivalent to inversion, where weaker edits are obtained by inverting up to timestep $n_{\max}$, and sampling from there. We illustrate the effect of $n_{\max}$ in App. C.

Algorithm 1 provides a simplified overview of our algorithm. For a more detailed version, please see Alg. S1.

5.2. Relation to optimization based methods

Superficially, FlowEdit may seem similar to optimization based methods like DDS [15] and PDS [22]. These techniques, introduced for diffusion rather than flow, attempt to minimize a loss that measures the alignment between the model's noise predictions with the source and target prompts. They approximate the gradient of the loss using only forward-passes through the model. For example, in DDS, the update step is the difference between the two noise predictions, which can be considered similar to our $V^{\Delta}_t(\hat{Z}^{\text{src}}_t, \hat{Z}^{\text{tar}}_t)$ direction. However, as we show in App. L, writing $V^{\Delta}_t(\hat{Z}^{\text{src}}_t, \hat{Z}^{\text{tar}}_t)$ in terms of noise predictions reveals that it is different from both the DDS and the PDS updates.

One may still wonder whether FlowEdit can be viewed as an optimization process, only for a different loss. Specifically, similarly to DDS, which attempts to minimize the norm of the difference between noise predictions in diffusion models, maybe FlowEdit attempts to minimize the norm of the difference between velocity predictions in flow

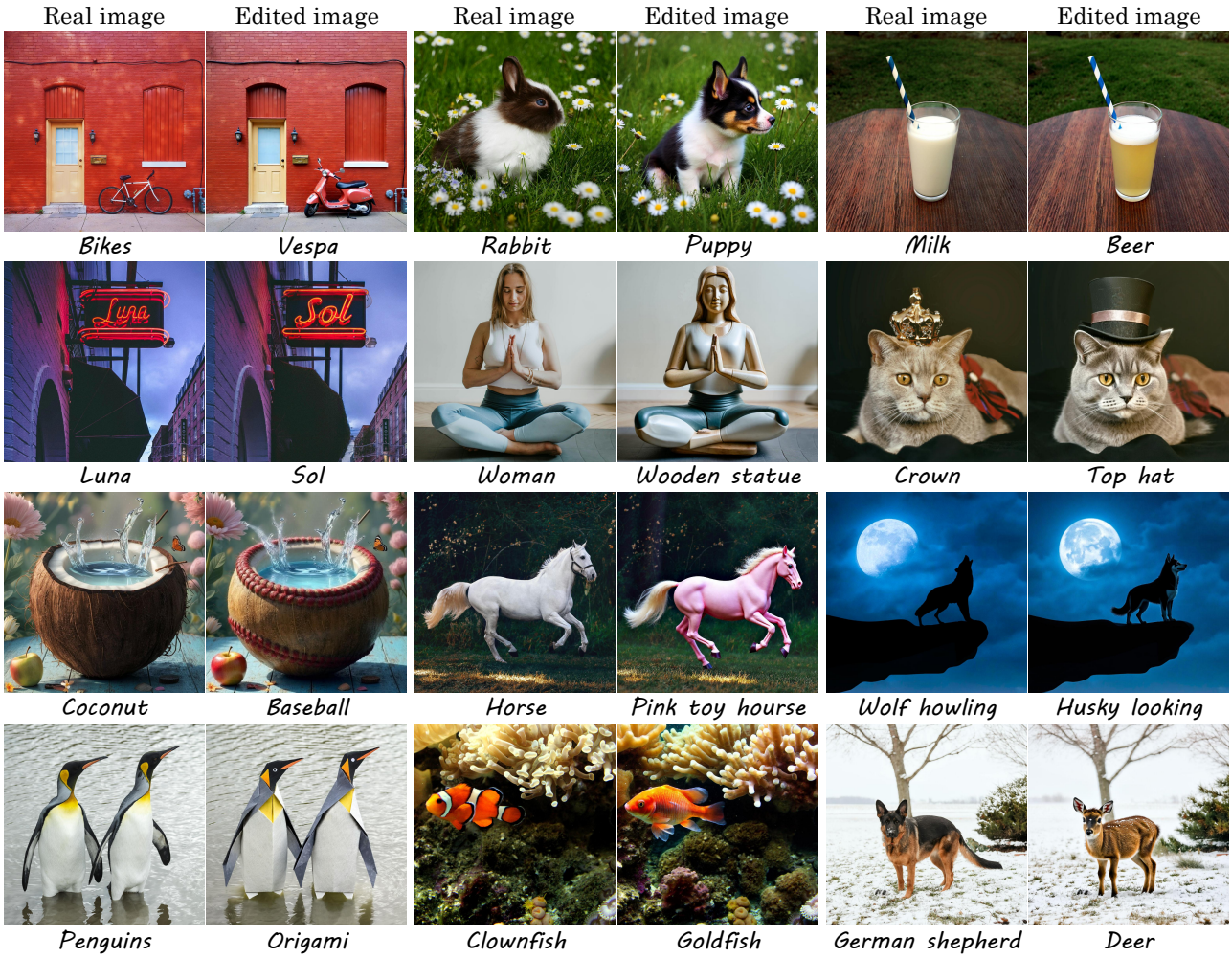


Figure 5. **FlowEdit results.** FlowEdit successfully edits diverse images using various editing prompts. The edits preserve the structure of the original image, changing only the specified region. FLUX was used for the first and third rows and SD3 for the second and fourth rows.

models? In App. L, we show that this optimization viewpoint is unjustified even for DDS. Namely, the DDS iterations do not decrease the DDS loss; they rather tend to *increase* it. Moreover, if allowed to continue beyond the default number of iterations, the quality of the edited image deteriorates. The optimization viewpoint is unnatural also because unlike standard optimizers, FlowEdit does not choose timesteps at random but rather uses a monotonically decreasing schedule $\{t_i\}_{i=0}^{n_{\max}}$. Additionally, it must use a learning rate of exactly $dt = t_{i-1} - t_i$ at iteration i (namely, the sum of all step-sizes must be 1). Any slight deviation from this learning rate causes the results to deteriorate significantly (see App. L).

5.3. Comparison to editing by (exact) inversion

To demonstrate the reduced transport cost of FlowEdit compared to editing-by-inversion, we evaluate both methods on a synthetic dataset of model-generated images. This way, the initial noise maps are known, ensuring the inversion is exact and eliminating potential issues of approximate inversion. The dataset consists of 1000 cat images generated by SD3 using variations of the prompt "a photo of cat" generated by Llama3 [9]. We edit these images using both FlowEdit and inversion, with the target prompt identical to the source prompt, except for replacing "cat" with "dog". We calculate the transport cost for both methods by measuring the MSE between source and edited images in the model's latent space, as well as LPIPS [50] on the decoded images. As expected, FlowEdit achieves a lower transport cost (1376 vs. 2239 for MSE, 0.15 vs. 0.25 for LPIPS), indicating superior preservation of the structure and semantics of the source image. Figure 4 shows a small qualitative comparison for the cat-to-dog edits.

To assess the alignment of our edits with the target distri-



Figure 6. **Qualitative comparisons.** With both SD3 (left) and FLUX (right), FlowEdit better adheres to the target prompt while simultaneously preserving the structure of the original image. The value next to SDEdit indicates the strength.

bution, we generated 1000 dog images with SD3, using the same target prompts but with "dog" instead of "cat". We followed by calculating FID [17] and KID [4] between the generated dog images and the edited dog images, for both methods. Our method achieves lower FID (51.14 vs. 55.88) and KID (0.017 vs. 0.023), indicating that similarly to inversion, our ODE path is able to produce images from the target distribution. See App. M for more details and results.

6. Experiments

Implementation details. In our experiments we use the official weights of SD3 medium [42] and FLUX.1 dev [25], available at HuggingFace, as the base T2I flow models. For SD3 we use $T = 50$ steps, with $n_{\max} = 33$. SD3 employs CFG [18] for their text conditioning. We set the source and target scales to 3.5 and 13.5, respectively. For FLUX we use $T = 28$ steps, with $n_{\max} = 24$. FLUX takes CFG as an input conditioning, which we set to the values of 1.5 for the source conditioning, and 5.5 for the target. For both methods we use $n_{\text{avg}} = 1$ (see Sec. 5.1 and App. K). We use these hyperparameters for all the results in Figs. 5,6,7.

Dataset. Our dataset consists of a diverse set of over 70 real images of dimension 1024^2 from the DIV2K dataset [1] and from royalty free online sources [35, 36]. Each image has a source prompt, which was obtained from LLaVA-1.5 [27] and manually refined. The source prompts have little effect on FlowEdit's results. In particular, they can be omitted (App. I). For each image, several handcrafted target prompts are provided to facilitate diverse edits. Overall, the dataset consists of over 250 text-image pairs, and is used to evaluate our and the competing methods. The dataset, including source and target prompt pairs, is available on our official github repository.

Competing methods. We compare our method against several competing text-based real image editing methods that use flow models. The first is editing by ODE inversion, which we apply with the same hyperparameters as our method. The second is SDEdit [30], which is easily applied to flow models by adding noise to the source image up to a specified $n_{\max}$ step and then performing regular sampling conditioned on a target prompt to obtain the edit. The parameter $n_{\max}$ controls the strength of the edit. Additionally, we compare to iRFDS [49], which is an SDS-based editing method for flow models. We use their official implementation and hyperparameters, which are available only for SD3. Furthermore, we compare to RF-Inversion [39]. As an official implementation is not available at the time of writing, we implement it based on the provided pseudo-code and the hyperparameters reported in the SM, which are available only for FLUX. Finally, we compare to RF Edit [46]

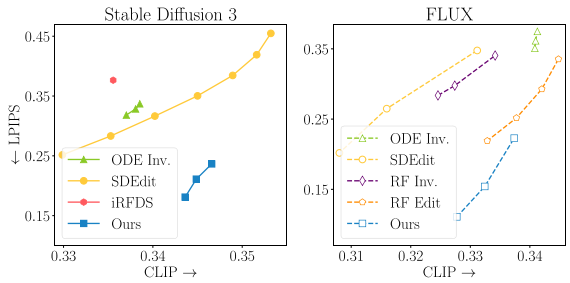


Figure 7. **Quantitative comparisons.** FlowEdit achieves a favorable balance between text adherence (CLIP) and structure preservation (LPIPS) compared to other methods. Connected markers represent different hyperparameters (see App. B).

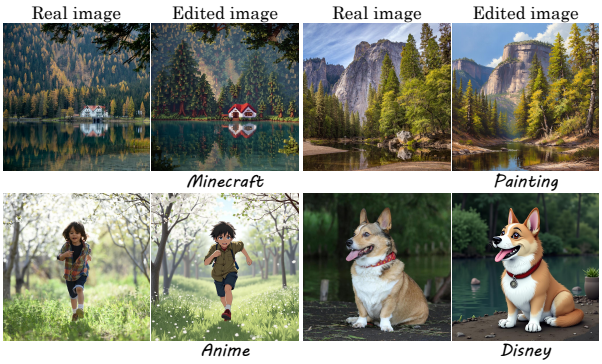


Figure 8. **Text-based style editing.** FlowEdit changes the style of an image at the cost of slightly deviating from the original image structure. SD3 was used for the first row and FLUX for the second. Hyperparameters used for these results are reported in App. D.

using their official implementation, which is also available only for FLUX. Whenever possible, we performed a hyperparameters search for each method to identify the optimal settings for editing. The final parameters used to create the edited images in the paper are detailed in App. B. We do not compare FlowEdit to text-based image editing techniques for diffusion models [6, 19, 44], as these are not easily adapted to flow models, and a direct comparison would not be fair due to model differences.

Qualitative evaluation. Figures 1, 5, S1 show editing results obtained with FlowEdit across a diverse set of images and target prompts, including localized edits to text and objects, modifications of multiple objects at once, pose changes, etc. Our edits exhibit good structural preservation of the source image, and simultaneously maintain good adherence to the target text. Figure 6 shows comparisons between FlowEdit and the competing methods for both SD3 and FLUX. FlowEdit is the only method that consistently adheres to the text prompt while preserving the structure of the original image. For example, FlowEdit is the only method that can both change the large text on the gas station sign (third row) and preserve the cars and background of the source image. Some methods, like RF-Inversion, are able to change the text on the sign but fail to maintain the structure of the original image, while others, such as iRFDS, somewhat preserve the structure but fail to fully change the sign. In the last row, while all methods perform the required edit (changing the decoration to strawberries), only FlowEdit (for both SD3 and FLUX) preserves the background accurately. Other methods introduce additional, unintended elements such as furniture and flowers. See App. B for more qualitative comparisons.

Quantitative evaluation. We numerically evaluate the results of FlowEdit and the other methods using LPIPS [50] to measure the semantic structure preservation (lower is better) and CLIP [38] to assess text adherence (higher is better). Figure 7 displays these results for SD3 (left) and FLUX (right), with varied hyperparameters. The plots show that FlowEdit achieves a favorable balance between structure preservation and text adherence. The other methods either maintain the structure of original image at the cost of a weak edit or modify the image with little regard to its original semantics. Additional quantitative evaluations and details, including additional similarity metrics beyond LPIPS, are provided in App. B.

Text-based style editing. The hyperparameters used for the previous experiments lead to strong structure preservation and are thus not optimal for style editing. Figure 8 demonstrates the results of FlowEdit for text-based style editing, where by allowing some deviation from the original structure, we gain stylistic flexibility. To achieve these results, we simply remove the dependence on the source image in the final generation steps, as discussed in App. D.

7. Conclusion and limitations

We introduced FlowEdit – an inversion-free, optimization-free and model agnostic method for text-based image editing using pre-trained flow models. Our approach constructs a direct ODE between the source and target distributions (corresponding to source and target text prompts), without passing through the standard Gaussian distribution as in inversion-based editing. Evaluations on synthetic datasets show that FlowEdit achieves lower transport costs, and thus stronger structure preservation. This translates to state-of-the-art performance across various editing tasks, as we illustrated with FLUX and SD3. While FlowEdit's strong structure preservation is beneficial for precise editing tasks, it can become a limitation when substantial modifications to large regions of the image are desired. This is illustrated in Fig. S6 in the context of pose and background editing.

Acknowledgements

This research was supported by the Israel Science Foundation (grant no. 2318/22) and by the Ollendorff Minerva Center, ECE faculty, Technion.

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